

Phase 1 — API Mastery (2 Weeks)

Goal: Understand how services communicate.

You should become comfortable with:

- REST APIs
- HTTP methods (GET, POST, PUT, DELETE)
- JSON data
- authentication keys
- webhooks

These are the backbone of services like OpenAI APIs and automation platforms like Zapier.

What to learn

Concepts

1. What an API is
 2. HTTP requests and responses
 3. JSON structure
 4. Authentication (API keys, headers)
 5. Webhooks
-

Best resources

Course

Search on YouTube:

“Python API Requests Tutorial”

or

“REST API Crash Course Python”

These are enough. Avoid long theoretical courses.

Documentation

Practice with real APIs like:

- OpenAI API
- Telegram Bot API

Reading documentation is a **core engineering skill**.

Mini projects (very important)

Build these during the two weeks:

1. **Weather API script**
Fetch weather data and print it.
 2. **Telegram bot**
Send messages automatically.
 3. **OpenAI text summarizer**
Send text → receive summary.
-

Daily schedule (4 hours)

1 hour — learning concepts
3 hours — coding experiments

Never reverse this ratio.

Phase 2 — Databases (2–3 Weeks)

Goal: Store and retrieve data.

Most AI systems fail without proper data storage.

Learn SQL using:

- PostgreSQL
or
- SQLite

SQLite is easier for beginners.

What to learn

Core SQL concepts:

- tables
- rows
- SELECT queries
- INSERT data
- UPDATE
- DELETE
- JOIN queries

These are **90% of real SQL usage**.

Best resources

Interactive learning:

SQLBolt

or

W3Schools SQL Tutorial

They are fast and practical.

Mini projects

1. Create **database of AI conversations**
Store prompts and responses.
2. Store **Telegram messages into database**
3. Build **simple analytics**
Count most common questions.

Now you are building **real data systems**.

Daily schedule

1 hour SQL learning

3 hours building scripts with database

Phase 3 — Backend Basics (3–4 Weeks)

Now you learn how to create your own APIs.

Use:

FastAPI

It is widely used in AI startups because it is fast and simple.

What to learn

Core concepts:

- API endpoints
 - request/response
 - routing
 - JSON responses
 - connecting database
 - calling AI APIs
-

Best resources

Search on YouTube:

“FastAPI Full Course”

Avoid huge frameworks like Django for now.

Mini projects

Build these systems:

Project 1

AI text summarization API

Input:

POST /summarize

Output:

summary from OpenAI.

Project 2

AI automation system

Pipeline:

Email → FastAPI → AI summary → store in database → Telegram alert

Now you are building **real automation pipelines**.

Phase 4 — Cloud (After 2 Months)

Only after building systems locally.

Then learn cloud platforms like:

- Amazon Web Services
- Google Cloud

Start with simple services:

- server deployment
- cloud functions
- API hosting
- database hosting

At this stage cloud will feel **logical instead of confusing**.

Recommended daily schedule (very important)

You have **4 hours/day**.

Use this structure.

Hour 1

Concept learning.

Hour 2

Follow tutorial while coding.

Hour 3

Build your own variation.

Hour 4

Debug, experiment, break things.

Learning happens during **debugging**, not watching.

Weekly output goal

Every week you must produce:

- 1 working script
- 1 small automation
- 1 GitHub project

Without output, progress is an illusion.

The real benchmark

After **8 weeks**, you should be able to build something like:

User request

↓

FastAPI backend

↓

AI model call

↓

Database storage

↓

Automation trigger

↓

Telegram notification

If you can build this system, you have moved from **learner** → **AI builder**.